

CASE STUDY

OISD/CS/2026-27/E&P/09

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INTRODUCTION

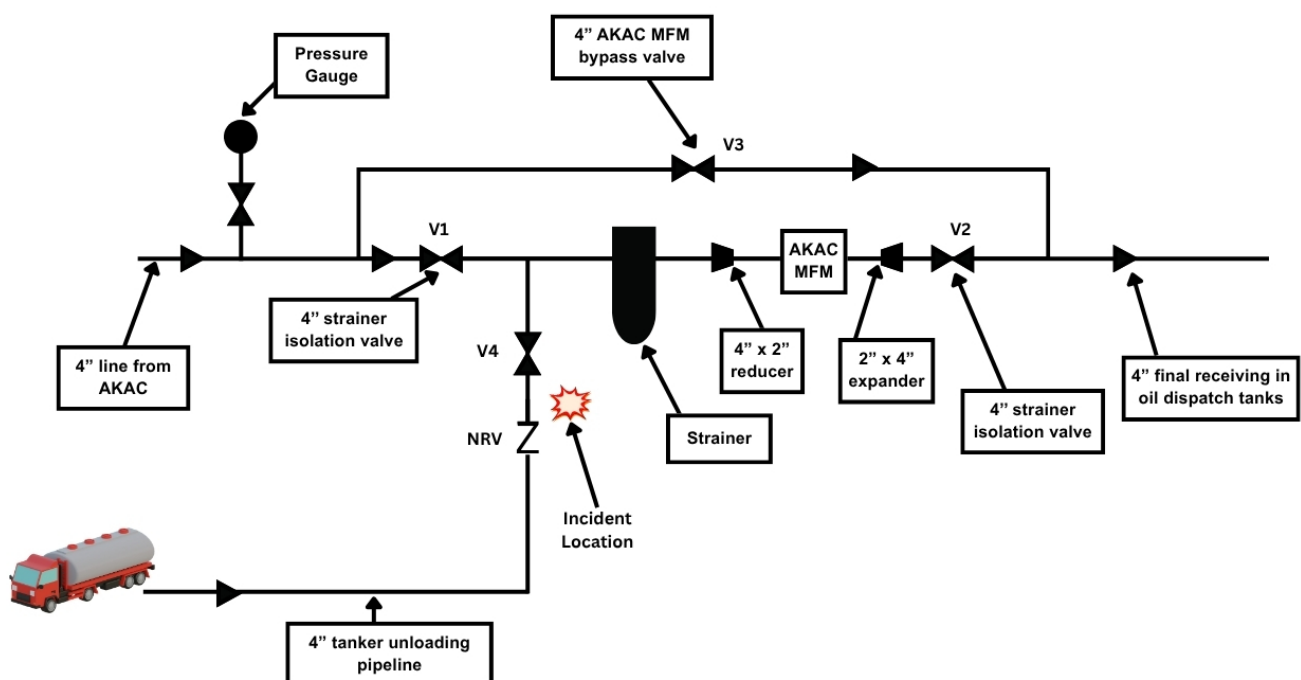
Title: Major incident at a GGS.
Location: Onshore
Loss/ Outcome: Loss of vision of a Contractual worker.

BRIEF OF INCIDENT

During pigging operation of pipeline from EPS to GGS, a severe blockage, caused by wax and sludge, led to a massive pressure spike against normal pressure of 6–10 kg/cm².

A drop-wise oil leak was observed at a 4-inch pipeline flange at GGS.

A four-member contractual crew isolated, drained, and depressurized the mass flow meter strainer section by valves to clean it, though no positive isolation (blinding) was performed. After clearing the pig receiver barrel—which contained heavy sludge but no pig—the crew started to box up the strainer. While three members worked on the strainer, the fourth crew member (IP) independently attempted to replace the leaking gasket on the tanker unloading pipeline. This section contained pressure trapped between a valve and NRV, not released during meter section depressurization. He loosened the top two studs of the flange. Trapped high-pressure crude oil and oily sludge suddenly burst through the gap and struck his right eye, causing a severe injury that resulted in a permanent loss of vision.



Provided for information purpose only. This information should be evaluated to determine if it is applicable in your operations, to avoid recurrence of such incidents.

OBSERVATIONS/ SHORTCOMINGS

1. Provision to ensure zero pressure after draining was not available between NRV and isolation valve V4 in pipeline.
2. It was evident from the position of the IP during work that he was not following the safe practice of avoiding standing in the direct line of any potential pressure discharge.
3. All the workers involved in the job were in the "unskilled" category. No supervision was available.
4. Appropriate PPE, such as safety goggles and safety helmets, was not used while carrying out the job of refitting the strainer.
5. For this job of gasket replacement, Work Permit, Job Safety Analysis (JSA), or Toolbox Talk (TBT) were not followed.

ROOT CAUSE OF THE INCIDENT

1. Non-confirmation of depressurization before opening flange.
2. Deficient leadership, supervision over activities & lack of co-ordination.
3. Inadequate training and competency in handling pressurized systems & isolation–depressurization procedures.
4. Weak safety culture regarding non-adherence to PTW system, JSA & PPE.

RECOMMENDATIONS

1. It should be ensured that the section between the Non-Return Valve (NRV) and isolation valve V4 of the pipeline has zero pressure.
2. Unauthorized maintenance/troubleshooting work should be strictly prohibited.
3. Conduct awareness & training sessions to ensure personnel maintain a safe position away from the direct line of any potential pressure discharge during work activities involving pressurized systems.
4. All workers involved in the work were unskilled workers. Supervision by responsible authorized personnel should be ensured.
5. Use of suitable PPE, especially eye protection, face protection, helmets, gloves, and other task-specific PPE should be ensured.
6. A formal Permit-to-Work (PTW) system should be followed. Maintenance or non-routine job should not be undertaken without approved work instructions, JSA, TBT, work permit documentation and supervision.
